

1. Anaphylaxis: management

Resuscitation Council UK (last updated 2021), NICE CG134 2011 (updated 2021), MHRA 2021 Drug Safety Update Vol 15(4), BNF (2020)



Red Whale

GEMS

Guidelines & Evidence Made Simple

1. ASSESSMENT

Assess: **A**irway, **B**reathing, **C**irculation, **D**isability, **E**xposure.

Anaphylaxis can be difficult to diagnose. Look for:

Acute onset

+

LIFE-THREATENING changes in

+/-

Skin changes:
flushing, urticaria, angioedema
(subtle or absent in 20%)

Airway

Swelling
Hoarseness
Stridor
Feels throat is closing up
Difficulty breathing/swallowing

Breathing

Increased respiratory rate
Wheeze
Fatigue
Cyanosis
Sats <94%

Circulation

Pale, clammy
Tachycardia
Feeling faint
Low BP
Confusion
Collapse

Skin changes with no compromise of airway/breathing/circulation are NOT enough to diagnose anaphylaxis.

GI symptoms may occur (vomiting, abdominal pain, faecal incontinence)

2. GET HELP

Summon other clinicians AND dial 999 for an ambulance

3. TREAT

Position and remove trigger

- If possible, remove trigger, e.g. sting.
- Position the patient comfortably, e.g. lying flat with legs raised if hypotensive; if airway or breathing difficulties, may be more comfortable semi-recumbent.
- Fatality can occur in minutes if a patient stands, walks or sits up suddenly. Patients must not walk or stand.

When skills/equipment available:

Establish airway

High-flow oxygen

Monitor: sats, BP, ECG

(do not delay treatment to attach monitoring)

If no response: fluid challenge

(iv crystalloid: adult 0.5–1litres, child 10ml/kg)

Asthma and anaphylaxis

Asthma and anaphylaxis symptoms can overlap and outcomes can be worse: see page 2

Administer adrenaline

Repeat after 5min if there is no improvement

**Give early:
do not delay!**

Adrenaline 1 in 1000 dose		Volume if 1mg/ml	
Adult (>12y)	500mcg	0.5ml	
Child 6–12y	300mcg	0.3ml	
Child 6m–6y	150mcg	0.15ml	
Child <6m	100–150mcg	0.1–0.15ml	

Give intramuscularly into anterolateral thigh

Do NOT give antihistamines/steroids

- Antihistamines only have a place once resuscitation is complete, and only then for skin reactions/angioedema. They have no place in resuscitation as they are ineffective. Oral non-sedating antihistamines are now preferred.
- Hydrocortisone is no longer recommended as part of the routine emergency management of anaphylaxis.

4. ADMIT

All need observation

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2. Anaphylaxis: drug issues

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Auto-injectors

Patients who need an auto-injector should:

- **Carry TWO auto-injectors at all times.** Discourage having 2 in the house, 2 in the handbag, 2 in the car. Why? If the devices are on your person, you don't need multiple sets of 2 devices. The Resuscitation Council says: "A carefree attitude may develop if multiple devices are prescribed." There may be times when more than 2 are needed, e.g. children who have two homes. Schools often ask for 2.
- **Know how to use it.** Devices are operated in different ways: make sure the patient knows how to use the type they have. Try to avoid switching brands. The NHS site has links to the patient videos for each brand: <https://www.nhs.uk/conditions/anaphylaxis/treatment/>
- **Know when to use it.** Give second dose 5 min after the first **if NOT starting to recover.**
- **Know to call an ambulance if the auto-injector is ever used, even if symptoms are improving** (NICE 2020 update, MHRA 2014).
- **At a medication review:** ask if they have their auto-injector on them (and if not, why not?!). Using a dummy device, ask the patient to demonstrate how to use it.

Device	150mcg dose		300mcg dose	500mcg dose
Emerade	Child up to 15kg	For children with body weights between these doses,  the BNF says can give 150mcg, or 300mcg may be more appropriate for some	Children from 30kg and adults	From age 12y
Epipen	Child up to 15kg		Children from 26kg and adults	Not available
Jext	Child up to 15kg		Children from 31kg and adults	Not available

Adrenaline administered by professionals

- In all healthcare settings, giving adrenaline from an ampule by syringe and needle is preferred in an emergency. Auto-injectors are not recommended in healthcare settings for administration of adrenaline in patients needing more than one dose.
- Use a blue needle except in very small infants (use orange).
- Inject at 90 degrees to the skin. Don't pinch the skin (risk goes subcut).
- ALWAYS give adrenaline intramuscularly: it is safer, easier and has greater margin for error than iv.

Drug dilemmas

ASTHMA and ANAPHYLAXIS

Death is more likely in those with poorly-controlled asthma, or in asthmatics who fail to use adrenaline early when developing anaphylaxis. In those with asthma, anaphylaxis can present with a spectrum of symptoms, ranging from:

Anaphylaxis symptoms WITHOUT asthma symptoms	Mixture of anaphylaxis and asthma symptoms	Asthma symptoms WITHOUT anaphylaxis symptoms
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If pure asthma symptoms are present, manage anaphylaxis AND treat according to the BTS guidelines for life-threatening asthma; otherwise, manage as for anaphylaxis alone.

BETA-BLOCKERS and ANAPHYLAXIS

- **Adrenaline works less well/may be ineffective in those on beta-blockers BUT do NOT give more than standard doses.** Glucagon is sometimes used on ITU to increase heart rate/contractility.
- **Think carefully before starting a beta-blocker in those at risk of anaphylaxis.**

3. Anaphylaxis: after an episode

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After an episode of anaphylaxis

- **Anaphylaxis can be biphasic so all patients should be admitted for observation** (at least 6h, up to 24h). They may be discharged with antihistamines and oral steroids for 3d to reduce urticaria and the risk of relapse. Steroid doses are not specified but consensus seems to be for between 30 and 60mg of prednisolone daily for adults. Before discharge, all should be offered an auto-injector as an interim measure before allergy clinic appointment. They should have had a brand-specific demonstration of when and how to use, and a prescription for 2 further auto-injectors to carry at all times.
- **All should be referred to a specialist allergy clinic** (list of NHS allergy clinics is available here: <https://www.bsaci.org/workforce/find-a-clinic/>). Specialist will decide who needs ongoing access to an auto-injector:
 - Usually recommended if known trigger that is difficult to avoid (stings, some foods) or idiopathic anaphylaxis.
 - NOT usually recommended for reactions to drugs (because easier to avoid).
- **After an episode of anaphylaxis, patient/family education is crucial. They must:**
 - Know what they are allergic to, what that allergen may be found in and how to avoid it.
 - Know how to recognise symptoms early and how to summon help.
 - Know how to use an auto-injector and to carry TWO at all times. The second dose should be administered 5 minutes after the first **if the person is not starting to recover**. If the auto-injector is ever used, even if symptoms are improving, an ambulance must be called.
 - **Consider using a MedicAlert bracelet.**

Is there a test to confirm anaphylaxis?

Mast cell tryptase can be measured after the patient has been stabilised if anaphylaxis is suspected. Plasma levels peak at 1–2h and drop rapidly thereafter, returning to normal within 6–8h. Three timed samples are needed. The first should be sent as soon as the patient is stable, the second 1–2h after onset and the third 24h after onset. Never delay resuscitation in order to obtain the blood sample! *We feel the hospital should make the decision about whether this is needed.*

Frequency, triggers and speed of onset

NICE suggests:

- About 1 in 1333 people will have an anaphylactic reaction in their lifetime.
- There are about 20 deaths per year in the UK from anaphylaxis, half of which are iatrogenic.
- Death is more likely in those with poorly-controlled asthma, or in asthmatics who fail to use adrenaline early when developing anaphylaxis.
- In many cases, no trigger is identified. For those that are, the commonest triggers are:
 - **Foods:** nuts being the most common food trigger (although in children, cow's milk is the commonest cause of fatal anaphylaxis (BMJ 2021;372:n251)).
 - **Drugs:** frequently antibiotics, NSAIDs, aspirin and anaesthetic muscle relaxants.
 - **Stings:** wasps and bees are the commonest in the UK.
- Food reactions are more common in children; drug reactions are more common in adults.
- Onset of reaction: shock and collapse usually occur within:
 - 5mins for intravenous drugs.
 - 10–15min for stings.
 - 30–35min for food.

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